

A beta partial least squares regression model: Diagnostics and application to mining industry data

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We propose a methodology based on partial least squares (PLS) regression models using the beta distribution, which is useful for describing data measured between zero and one. The beta PLS model parameters are estimated with the maximum likelihood method, whereas a randomized quantile residual and the generalized Cook and Mahalanobis distances are considered as diagnostic methods. A simulation study is provided for evaluating the performance of these diagnostic methods. We illustrate the methodology with real-world mining data. The results obtained in this study based on the beta PLS model and its diagnostics may be of interest for the mining industry.

KEYWORDS

Cook distance, generalized linear models, Mahalanobis distance, NIR spectra data, principal component analysis, quantile residual, R software

1 | INTRODUCTION

Partial least squares (PLS) regression is a multivariate technique that was developed for data modeling when multicollinearity in explanatory variables (covariates) is present. PLS regression is also used when the number of parameters in the model is greater than the number of cases (data); see Hair et al.¹ The PLS model combines multiple regression and principal component analysis (PCA). The PLS regression is a method that reduces the amount of covariates to a smaller set of uncorrelated components, called PLS components and corresponding to latent factors. The number of selected PLS components allows us to maximize the covariance between the response variable and the covariates considered. To obtain the PLS components, an optimization method is performed by means of ordinary least squares (OLS).

The PLS regression was introduced by Wold,² who showed that if the same number of PLS components and covariates is selected then the estimated regression coefficients, and their corresponding standard errors (SEs), coincide with those obtained by OLS. Although according to the logic of the least squares method, PLS regression should be free of any distributional assumption, normality is often assumed. This is because, under normality, least squares and maximum likelihood (ML) estimates of the PLS coefficients coincide. We refer to the PLS model based on the normal distribution as normal PLS regression in short. Normal PLS models are widely used in bioinformatics, biology, chemometrics, and medicine, among other areas; see previous studies.^{3–6} Assuming normality in PLS models opens the possibility to consider there other distributions. In particular, an extension of the normal PLS regression was proposed by Marx⁷ on the basis of generalized linear models (GLM), which we name PLS-GLM in short; see McCullagh and Nelder⁸ and Bastien et al.⁹

The beta distribution has been largely used for modeling data measured between zero and one. This distribution has good properties, is flexible, and can be used in a wide range of applications; see Johnson et al.¹⁰ and Kotz and van Dorp¹¹

for more details on the beta distribution. Bertrand et al.⁶ proposed an extension of the normal PLS regression using the beta distribution and considering PLS-GLM, which we name beta PLS regression in short. To optimally find the number of PLS components, in normal PLS regression or PLS-GLM, the prediction error sum of squared (PRESS) method is often used, which is a cross-validation measure; see Li et al.¹² Other methods used for finding the number of PLS components in PLS-GLM focus on information criteria, which are based on ML methods; see Bastien et al.⁹

Diagnostic analysis is a necessary aspect to be considered in all statistical modeling, once the estimation of parameters has been performed. This analysis is conducted to assess the suitability of the distributional assumption and the sensitivity and stability of the parameter estimation. Diagnostics can be performed by residuals and global influence methods, such as the generalized Cook and Mahalanobis distances. Residuals allow us to detect the distributional assumption and identify atypical cases, which may affect the results from the statistical analysis of data. Global influence methods remove cases and evaluate their effect on the fitted model globally; see Cook and Weisberg¹³ and Chatterjee and Hadi.¹⁴ For the use of residuals and other diagnostic methods in nonnormal models, including beta regression, see, for example, previous studies.¹⁵⁻²³

Kaolinite is a clay mineral, which is part of the group of industrial silicate minerals, collected in mining; see Orbovic and Huang.²⁴ Kaolinite is present in rock samples obtained by batch, and it is often measured by near-infrared (NIR) spectrometry with wavelengths between 350 and 2500 nm with 200 scans per sample in 20 seconds; see Burns and Ciurczak.²⁵ Data related to kaolinite are often measured between zero and one as a proportion. Then, these data can be adequately described by the beta distribution. However, proportion of kaolinite is explained by spectral variables, which usually correspond to a high number of covariates many of them correlated. Thus, a beta PLS regression should be used to model the proportion of kaolinite in function of spectral variables instead of a normal PLS regression.

The main objective of this paper is to propose a methodology based on beta PLS regression to predict the proportion of kaolinite using NIR measurements. Specifically, in the proposed methodology, the PLS coefficients are estimated by the ML method. Hence, diagnostic methods considering the quantile residual and the generalized Cook and Mahalanobis distances are derived, which allow us to evaluate the suitability of the beta PLS regression and to detect atypical data. A simulation study is conducted for evaluating the performance of the diagnostic methods considered in this study. The methodology is applied to real-world mining data and compared with the existing models in the literature to predict data related to the proportion of kaolinite using NIR spectral data as covariates. In the application, the parameters of each competing model are estimated, and then the best model is selected. Consequently, the optimal number of PLS components is obtained for the best model and its diagnostic analysis is performed.

This research work is organized as follows. In Section 2, modeling aspects are introduced. In Section 3, diagnostic methods for the beta PLS regression model are derived. In Section 4, the application of the beta PLS regression and its diagnostics to predict the proportion of kaolinite is presented, using NIR spectral data in a mining problem. Also in Section 4, we evaluate the performance of the diagnostic methods by simulation. In Section 5, the conclusions of this work are discussed.

2 | MODELING

2.1 | A beta regression model

The usual parameterization of the beta distribution considers the shape parameters a and b for the random variable Y to be modeled; see Kotz and van Dorp.¹¹ This is denoted by $Y \sim \text{Beta}(a, b)$, and the corresponding probability density function (PDF) is given by

$$f(y; a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} y^{a-1} (1-y)^{b-1}, \quad 0 < y < 1, \quad a > 0, \quad b > 0, \quad (2.1)$$

where Γ is the standard gamma function. Then, the mean and variance of Y are expressed as

$$E[Y] = \frac{a}{a+b}, \quad \text{Var}[Y] = \frac{ab}{(a+b+1)(a+b)^2}, \quad (2.2)$$

respectively. Ferrari and Cribari-Neto²⁶ proposed a reparameterization of the beta distribution based on its mean defined in (2.2), denoted now by μ , and a precision parameter δ , both given by

$$\mu = \frac{a}{a+b}, \quad \delta = (a+b). \quad (2.3)$$

From (2.3), note that $a = \mu\delta$ and $b = (1 - \mu)\delta$. By using the reparameterization assumed in (2.3), denoted here by $Y \sim \text{Rbeta}(\mu, \delta)$, we have that the mean and variance of Y are now expressed as

$$E[Y] = \mu, \quad \text{Var}[Y] = \frac{\mu(1 - \mu)}{1 + \delta}, \quad (2.4)$$

respectively. Note that the variance of Y given in (2.4) is a function of μ . Therefore, such as the gamma and reparameterized Birnbaum-Saunders distributions (see previous studies^{19,22,27}), the reparameterized beta distribution allows us to model heteroscedasticity. Under this reparameterization, we are able to model data by regression based on GLM and the beta distribution as detailed below; see also Ferrari and Cribari-Neto.²⁶

Let $Y = (Y_1, \dots, Y_n)^\top$ be a sample (independent random variables) of a response $Y \sim \text{Rbeta}(\mu, \delta)$, and $y = (y_1, \dots, y_n)^\top$ be a column vector with the observations of Y . Then, the mean of the response Y_i, μ_i namely, can be written as a beta regression model of GLM type of the form

$$E[Y_i] = \mu_i = g^{-1}(\underline{x}_i^\top \underline{\beta}), \quad i = 1, \dots, n, \quad (2.5)$$

where $\underline{x}_i^\top \underline{\beta} = \eta_i$ is a linear predictor, g^{-1} is the inverse function of the link function g , which must be twice differentiable subject to the interval $[0, 1]$, $\underline{\beta} = (\beta_0, \beta_1, \dots, \beta_k)^\top$ is a vector of unknown parameters to be estimated, and $\underline{x}_i^\top = (1, x_{i1}, \dots, x_{ik})$ are the observations of k covariates $\underline{X}^\top = (X_1, \dots, X_k)$, including the constant term. Note that $\underline{x}_i^\top$ corresponds to the i -th row vector of the design matrix

$$\underline{X} = \begin{pmatrix} 1 & x_{11} & \dots & x_{1k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{nk} \end{pmatrix}. \quad (2.6)$$

Observe that, as mentioned, although PLS regression is related to least squares, whose method does not need a distributional assumption for the response, in this work we assume a parametric distribution. Thus, the parameter estimation of the model defined in (2.5) is performed by the ML method. Then, on the basis of n observations $\underline{y}$ from the beta regression given in (2.5), the log-likelihood function for $\underline{\theta} = (\underline{\beta}^\top, \delta)^\top$ is expressed as

$$\begin{aligned} \ell(\underline{\theta}) = & \sum_{i=1}^n \log(\Gamma(\delta)) - \sum_{i=1}^n \log(\Gamma(\mu_i \delta)) - \sum_{i=1}^n \log(\Gamma((1 - \mu_i) \delta)) \\ & + \sum_{i=1}^n (\mu_i \delta - 1) \log(y_i) + \sum_{i=1}^n ((1 - \mu_i) \delta - 1) \log(1 - y_i), \end{aligned} \quad (2.7)$$

where $\mu_i = g^{-1}(\eta_i)$, with $\eta_i = \underline{x}_i^\top \underline{\beta}$ being defined in (2.5). The corresponding score vector, obtained by differentiating (2.7) with respect to $\underline{\theta}$, is given by $\underline{\dot{\ell}}(\underline{\theta}) = \partial \ell(\underline{\theta}) / \partial \underline{\theta} = (\underline{\dot{\ell}}_{\underline{\beta}}(\underline{\theta})^\top, \dot{\ell}_{\delta}(\underline{\theta}))^\top$, where

$$\underline{\dot{\ell}}_{\underline{\beta}}(\underline{\theta}) = \delta \underline{X}^\top \underline{F}(\underline{y}^* - \underline{\mu}^*),$$

with $\underline{X}$ defined in (2.6), $\underline{F} = \text{diag}\{1/g'(\mu_1), \dots, 1/g'(\mu_n)\}$, $g'(\mu_i) = dg(\mu_i)/d\mu_i$, $\underline{y}^* = (y_1^*, \dots, y_n^*)^\top$, and $\underline{\mu}^* = (\mu_1^*, \dots, \mu_n^*)^\top$, while

$$\dot{\ell}_{\delta}(\underline{\theta}) = \sum_{i=1}^n (\mu_i(y_i^* - \mu_i^*) + \log(1 - y_i) - \psi((1 - \mu_i)\delta) + \psi(\delta)),$$

with $y_i^* = \log(y_i/(1 - y_i))$, $\mu_i^* = \psi(\mu_i \delta) - \psi((1 - \mu_i)\delta)$, and ψ being the digamma function defined as $\psi(z) = d \log(\Gamma(z))/dz$. The ML estimates of $\underline{\beta}$ and δ , $\hat{\underline{\beta}}$ and $\hat{\delta}$ namely, are the solution to the system of equations given by $\underline{\dot{\ell}}_{\underline{\beta}}(\underline{\theta}) = \underline{0}$ and $\dot{\ell}_{\delta}(\underline{\theta}) = 0$, where $\underline{0}$ is a vector of zeros of a suitable dimension. Because the corresponding ML estimates cannot be expressed in a closed form, we compute them by maximizing the log-likelihood function defined in (2.7) numerically with a quasi-Newton algorithm; see more details on nonlinear optimization in Nocedal and Wright.²⁸

Ferrari and Cribari-Neto²⁶ showed that the second derivatives of $\ell(\theta)$ defined in (2.7), with respect to β and δ , are expressed as

$$\begin{aligned}\frac{\partial^2 \ell(\underline{\theta})}{\partial \beta_j \partial \beta_k} &= \sum_{i=1}^n \left(\frac{\partial^2 \ell_i(\mu_i, \delta)}{\partial \mu_i^2} \frac{d\mu_i}{d\eta_i} + \frac{\partial \ell_i(\mu_i, \delta)}{\partial \mu_i} \frac{\partial}{\partial \mu_i} \frac{d\mu_i}{d\eta_i} \right) \frac{d\mu_i}{d\eta_i} x_{ij} x_{ik}, \\ \frac{\partial^2 \ell(\underline{\theta})}{\partial \beta_j \partial \delta} &= \sum_{i=1}^n \left(y_i^* - \mu_i^* - \delta \frac{d\mu_i^*}{d\delta} \right) \frac{1}{g'(\mu_i)} x_{ij}, \\ \frac{\partial^2 \ell(\underline{\theta})}{\partial \delta^2} &= \sum_{i=1}^n (\psi'((1 - \mu_i)\delta)(1 - \mu_i) + \psi'(\delta)),\end{aligned}$$

where ψ' is the trigamma function defined as $\psi'(z) = d^2 \log(\Gamma(z))/dz^2 = d\psi(z)/dz$. Consequently, the Hessian matrix is given by

$$\ddot{\ell}(\underline{\theta}) = \begin{pmatrix} \frac{\partial^2 \ell(\underline{\theta})}{\partial \beta \partial \beta^\top} & \frac{\partial^2 \ell(\underline{\theta})}{\partial \beta \partial \delta} \\ \frac{\partial^2 \ell(\underline{\theta})}{\partial \delta \partial \beta^\top} & \frac{\partial^2 \ell(\underline{\theta})}{\partial \delta^2} \end{pmatrix}.$$

In addition, the expected Fisher information matrix is obtained as

$$\mathbf{K}(\underline{\theta}) = -E[\ddot{\ell}(\underline{\theta})] = \begin{pmatrix} \delta \mathbf{X}^\top \mathbf{B} \mathbf{X} & \mathbf{X}^\top \mathbf{F} \mathbf{v} \\ \mathbf{v}^\top \mathbf{F}^\top \mathbf{X} & \text{tr}(\mathbf{D}) \end{pmatrix}, \quad (2.8)$$

where $\mathbf{B} = \text{diag}\{b_1, \dots, b_n\}$, $\mathbf{v} = (v_1, \dots, v_n)^\top$, and $\mathbf{D} = \text{diag}\{d_1, \dots, d_n\}$, with

$$\begin{aligned}b_i &= \delta(\psi'(\mu_i\delta) + \psi'((1 - \mu_i)\delta))/(g'(\mu_i))^2, \\ v_i &= \delta(\psi'(\mu_i\delta)\mu_i - \psi'((1 - \mu_i)\delta)(1 - \mu_i)), \\ d_i &= \psi'(\mu_i\delta)\mu_i^2 + \psi'((1 - \mu_i)\delta)(1 - \mu_i)^2 - \psi'(\delta),\end{aligned}$$

and $\text{tr}(\mathbf{D})$ denoting the trace of $\mathbf{D}$. Note that the parameters β and δ are not orthogonal, in contrast to what is verified in the class of GLM; see McCullagh and Nelder.⁸ Usual regularity conditions (see Cox and Hinkley²⁹) are fulfilled for the beta PLS regression model if its parameters are within the parameter space. Then, the ML estimator $\hat{\underline{\theta}}$ is consistent and follows an asymptotic joint distribution, which is normal with asymptotic mean $\underline{\theta}$, and an asymptotic variance-covariance matrix $\Sigma(\underline{\theta})$. Thus, as $n \rightarrow \infty$ and recalling that $\underline{\theta} = (\beta^\top, \delta)^\top$, we have that

$$\sqrt{n}(\hat{\underline{\theta}} - \underline{\theta}) \xrightarrow{D} N_{k+2}(\mathbf{0}, \Sigma(\underline{\theta})), \quad (2.9)$$

where $\xrightarrow{D}$ denotes convergence in distribution. Note that if $\mathbf{J}(\underline{\theta}) = \lim_{n \rightarrow \infty} (1/n) \mathbf{K}(\underline{\theta})$ exists and is nonsingular, with $\mathbf{K}(\underline{\theta})$ being the expected Fisher information matrix given in (2.8), then $\Sigma(\underline{\theta}) = \mathbf{J}(\underline{\theta})^{-1}$. The diagonal elements of $\mathbf{K}(\underline{\theta})^{-1}$, $k_{jj}^{-1}(\underline{\theta})$ namely, may be used for approximating the corresponding asymptotic SEs, that is, by using

$$\text{SE}[\hat{\theta}_j] = \sqrt{k_{jj}^{-1}(\underline{\theta})}, \quad j = 1, \dots, k+2. \quad (2.10)$$

Note that $\hat{\mathbf{K}}(\underline{\theta})^{-1} = \mathbf{K}(\hat{\underline{\theta}})^{-1}$ is a consistent estimator of $\Sigma(\underline{\theta})$ and then the associated asymptotic SEs given in (2.10) may be estimated as $\widehat{\text{SE}}[\hat{\theta}_j] = (k_{jj}^{-1}(\hat{\underline{\theta}}))^{1/2}$, for $j = 1, \dots, k+2$. Asymptotic inference on the beta regression parameters can be conducted using (2.9); for more details, see Ferrari and Cribari-Neto²⁶ and Cribari-Neto and Queiroz.³⁰

2.2 | A normal PLS regression model

As mentioned, PLS regression is a technique alternative to OLS regression, which is particularly useful when the covariates are correlated. This correlation induces a multicollinearity problem, which appears with frequency in chemometrical applications of regression models. In addition, PLS regression is used when the amount of covariates is greater than the amount of cases ($k > n$). As also mentioned, PLS regression mixes multiple regression and PCA, acting similarly to PCA regression; see Jolliffe.³¹ However, unlike PCA regression, where the latent factors maximize the variance of the covariates, PLS regression extracts a set of latent factors that maximizes the covariance between the response and covariates.

Then, as in multiple linear regression, PLS regression is based on the linear model

$$\underline{Y} = \underline{X}\underline{\beta} + \underline{\varepsilon}, \quad (2.11)$$

where $\underline{Y}$ is the vector of responses, $\underline{X}$ is a design matrix containing the values of covariates, $\underline{\beta}$ is the vector of regression coefficients to be estimated, and $\underline{\varepsilon}$ is the vector of model errors. Before performing PLS regression, we must center the observed values of the covariates by subtracting their sample means, $\bar{x}_j$ namely, and then we must scale them by dividing their sample standard deviations (SDs), s_{X_j} say, that is, we work with

$$\underline{X}^* = \begin{pmatrix} \frac{x_{11} - \bar{x}_1}{s_{X_1}} & \dots & \frac{x_{1k} - \bar{x}_k}{s_{X_k}} \\ \vdots & \ddots & \vdots \\ \frac{x_{n1} - \bar{x}_1}{s_{X_1}} & \dots & \frac{x_{nk} - \bar{x}_k}{s_{X_k}} \end{pmatrix} = \begin{pmatrix} x_{11}^* & \dots & x_{1k}^* \\ \vdots & \ddots & \vdots \\ x_{n1}^* & \dots & x_{nk}^* \end{pmatrix}, \quad (2.12)$$

Thus, PLS regression produces a matrix of factor scores as linear combinations of the centered and scaled original covariates defined in (2.12). This matrix is given by

$$\underline{T} = \underline{X}^* \underline{W}, \quad (2.13)$$

where

$$\underline{T} = \begin{pmatrix} t_{11} & \dots & t_{1c} \\ \vdots & \ddots & \vdots \\ t_{n1} & \dots & t_{nc} \end{pmatrix}, \quad \underline{W} = \begin{pmatrix} w_{11} & \dots & w_{1c} \\ \vdots & \ddots & \vdots \\ w_{k1} & \dots & w_{kc} \end{pmatrix}$$

are factor (or PLS components) and weight matrices, respectively. Note that $c \leq \text{rk}(\underline{X}) \leq k$, with $\text{rk}(\underline{X})$ denoting the rank of the matrix $\underline{X}$ and c representing the optimal number of PLS components to be selected suitably, solving the high-dimensionality problem; see Section 2.4. All coefficients of correlation between the columns of the PLS component matrix $\underline{T}$ are zero, doing these columns to be orthogonal, solving the multicollinearity problem. Thus, PLS regression, which connects the linear model defined in (2.11) to the PLS components, is expressed as

$$\underline{Y} = \underline{X}^* \underline{\beta}_{\text{PLS}} + \underline{\varepsilon} = \underline{X}^* \underline{W} \underline{q} + \underline{\varepsilon} = \underline{T} \underline{q} + \underline{\varepsilon}, \quad (2.14)$$

where $\underline{Y}$ and $\underline{\varepsilon}$ are given in (2.11) and $\underline{X}^*$, $\underline{W}$, $\underline{T}$ are defined in (2.13), whereas $\underline{\beta}_{\text{PLS}}$ is the vector of PLS coefficients and $\underline{q} = (q_1, \dots, q_c)^T$ is a loading vector corresponding to the PLS coefficients. Once $\underline{q}$ is estimated, expression in (2.14) can be used as a predictive regression model. To compute the first PLS component, $t_1 = \underline{X}^* \underline{w}_1$ namely, the weight

$$\text{Cov}[Y, X_j] = \text{Cor}[Y, X_j] \text{SD}[Y] \text{SD}[X_j], \quad j = 1, \dots, p, \quad (2.15)$$

for the covariate X_j , must be determined, where $\text{Cov}[Y, X_j]$, $\text{Cor}[Y, X_j]$, $\text{SD}[Y]$, and $\text{SD}[X_j]$ are the covariance, correlation coefficient, and SDs of Y and X_j , respectively. Then, the covariate X_j is relevant for constructing t_1 if X_j and Y are highly correlated. Note that the weight $\text{Cov}[Y, X_j]$ defined in (2.15) is also the regression coefficient a_{1j} in an OLS regression of Y on the modified covariate $X_j/\text{Var}[X_j]$, that is,

$$a_{1j} = \frac{\text{Cov}[Y, X_j/\text{Var}[X_j]]}{\text{Var}[X_j/\text{Var}[X_j]]} = \frac{\text{Cov}[Y, X_j]/\text{Var}[X_j]}{\text{Var}[X_j]/(\text{Var}[X_j])^2} = \text{Cov}[Y, X_j], \quad (2.16)$$

which allows us to generate the vector $\underline{a}_1 = (a_{11}, \dots, a_{1k})^T$, which must be normalized. Thus, before obtaining the second PLS component, t_2 namely, a further regression must be generated as

$$\underline{X}_j^* = p_{1j} t_1 + \underline{\zeta}_j, \quad j = 1, \dots, k, \quad (2.17)$$

where $\underline{X}_j^*$ is associated with the j -th column vector of the centered design matrix given in (2.12), t_1 is the first PLS component, p_{1j} is an OLS regression coefficient to be estimated, and $\underline{\zeta}_j$ is the corresponding vector of model errors. Note that the vector $\underline{p}_{\underline{1}}$ is composed of the k scalar elements p_{1j} . Iteratively, the second PLS component is obtained by means of the vector $\underline{a}_2 = (a_{21}, \dots, a_{2k})^T$ related to the regression of Y on (T_1, X_j^*) , which must be normalized, and so on for the other components; see Bastien et al.⁹ For practical use, the second component t_2 must be expressed in terms of the covariate X_j^* . This is possible because, when determining t_2 , in addition to the regression of Y on (T_1, X_j^*) , the regression of X_j^* on (T_1, X_j^*) is computed, permitting the residuals $\underline{x}_{ij}^* = x_{ij}^* - \hat{p}_{1j} t_1$, obtained from (2.17), to be a function of the values x_{ij}^* of X_j^* , for $i = 1, \dots, n$. Consequently, maximizing the vectors $\underline{a}_1, \dots, \underline{a}_c$ defined as in (2.16), the covariance between Y and X_j is maximized and then the PLS regression model is established. Note that the PLS regression works in a different way

to the PCA regression, where, as mentioned, the PCA latent factors maximize the variance of the covariates; for more details on parameter estimation methods and algorithms in PLS regression, see previous studies.^{7,9,32} The PLS approach is summarized in Algorithm 1.

Algorithm 1 PLS approach

- 1: Start $\mathbf{X}_0^* = \mathbf{X}^*$ and $\underline{y}_0 = \underline{y}$.
 - 2: Calculate the first PLS component $\underline{t}_1$ following the steps:
 - 2.1 Estimate the coefficient a_{1j} of x_j^* in the OLS regression of Y on X_j^* , repeating this step for $j = 1, \dots, k$ and so constructing the vector $\underline{a}_1 = (a_{11}, \dots, a_{1k})^\top$.
 - 2.2 Determine the first column vector of $\mathbf{W}$ corresponding to the direction of maximum covariance as $\underline{w}_1 = (1/||\underline{a}_1||)\underline{a}_1$.
 - 2.3 Compute the first PLS component as $\underline{t}_1 = \mathbf{X}_0^* \underline{w}_1$.
 - 3: Calculate the second PLS component $\underline{t}_2$ following the steps:
 - 3.1 Estimate the coefficient a_{2j} of x_j^* in the OLS regression of Y on (T_1, X_j^*) , repeating this step for $j = 1, \dots, k$ and so constructing the vector $\underline{a}_2 = (a_{21}, \dots, a_{2k})^\top$.
 - 3.2 Determine the second column vector of $\mathbf{W}$ corresponding to the direction of maximum covariance as $\underline{w}_2 = (1/||\underline{a}_2||)\underline{a}_2$.
 - 3.3 Establish the residual matrix $\mathbf{X}_1^* = \mathbf{X}^* - \underline{t}_1 \hat{\underline{p}}_1^\top$, where each column of the matrix $\mathbf{X}_1^*$ corresponds to the residuals in the OLS regression of X_j^* on T_1 given in (2.17).
 - 3.4 Compute the second PLS component as $\underline{t}_2 = \mathbf{X}_1^* \underline{w}_2$.
 - 4: State the optimal number $c < k$ of PLS components and calculate the other PLS components $\underline{t}_h$, for $h = 3, \dots, c$, following the steps:
 - 4.1 Estimate the coefficient a_{hj} of x_j^* in the OLS regression of Y on $(T_1, \dots, T_{h-1}, X_j^*)$, repeating this step for $j = 1, \dots, c$ and so constructing the vector $\underline{a}_h = (a_{h1}, \dots, a_{hc})^\top$.
 - 4.2 Determine the h -th column vector of $\mathbf{W}$ corresponding to the direction of maximum covariance as $\underline{w}_h = (1/||\underline{a}_h||)\underline{a}_h$.
 - 4.3 Establish the residual matrix $\mathbf{X}_{h-1}^* = \mathbf{X}_{h-2}^* - \underline{t}_{h-1} \hat{\underline{p}}_{h-1}^\top$, where each column of the matrix $\mathbf{X}_{h-1}^*$ corresponds to the residuals in the OLS regression of X_j^* on T_{h-1} .
 - 4.4 Compute the h -th PLS component as $\underline{t}_h = \mathbf{X}_{h-1}^* \underline{w}_h$.
 - 5: Complete the matrices $\mathbf{T}$ and $\mathbf{W}$ with the vectors obtained in steps 2, 3 and 4.
 - 6: Complete the matrices $\mathbf{T}$ and $\mathbf{W}$ with the vectors obtained in steps 2, 3, and 4.
 - 7: Find the OLS estimate (with no intercept) of the loading vector $\underline{q}, \hat{\underline{q}} = (\hat{q}_1, \dots, \hat{q}_c)^\top$ namely, based on the PLS model given in (2.14) using the matrix $\mathbf{T}$, that is, $\hat{\underline{q}} = (\mathbf{T}^\top \mathbf{T})^{-1} \mathbf{T}^\top \underline{y}$.
 - 8: Compute the PLS coefficients using latent factors as $\hat{\underline{\beta}}_{\text{PLS}} = \mathbf{W} \hat{\underline{q}}$.
 - 9: Predict values for the response Y with the estimated PLS model given by $\hat{y}_{\text{pred}} = \underline{x}_{\text{pred}}^\top \hat{\underline{\beta}}$, where $\underline{x}_{\text{pred}}$ is a vector with the values of the covariates to be used for predicting and $\hat{\underline{\beta}}$ is associated with the PLS coefficients of the model $\hat{y} = \underline{x}^* \hat{\underline{\beta}}_{\text{PLS}}$ depending on the original variables, after the corresponding algebraic manipulation.
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2.3 | A beta PLS regression model

A beta PLS regression model of GLM type can be obtained from (2.5) and (2.14) as

$$E[Y_i] = \mu_i = g^{-1}(\underline{x}_i^{*\top} \mathbf{W} \underline{q}) = g^{-1}(\underline{t}_i^\top \underline{q}), \quad i = 1, \dots, n, \quad (2.18)$$

where μ_i and g^{-1} are defined in (2.4), whereas $\underline{t}_i^\top = (t_{i1}, \dots, t_{ic})$ is the i -th column vector of $\mathbf{T}$ and $\underline{x}_i^{*\top}$ is the i -th column vector of $\mathbf{X}^*$ given in (2.12). Algorithm 2 allows us to find the column vectors of $\mathbf{T}$ corresponding to the PLS components $\underline{t}_h = (t_{1h}, \dots, t_{nh})^\top$, for $h = 1, \dots, c$. Then, the predictive beta PLS regression model is obtained; see Bertrand et al.⁶ In Algorithm 2, note that the coefficients a_{1j} given in (2.16) cannot now be established in an analytical closed form and a numerical procedure, as a quasi-Newton algorithm, must be used on the basis of ML estimation produced from the likelihood function defined in (2.7).

Algorithm 2 Getting a predictive beta-PLS regression model

- 1: Start $\mathbf{X}_0^* = \mathbf{X}^*$ and $\underline{y}_0 = \underline{y}$.
- 2: Calculate the first PLS component t_1 following the steps:
 - 2.1 Estimate the coefficient a_{1j} of x_j^* in the beta-PLS regression of Y on X_j^* , using the ML method based on (2.7) and (2.18), where $\mu_i = g^{-1}(a_{1j}x_{ji}^*)$, with $i = 1, \dots, n$, repeating this step for $j = 1, \dots, k$ and so constructing the vector $\underline{a}_1 = (a_{11}, \dots, a_{1k})^\top$.
 - 2.2 Proceed as in steps 2.2 and 2.3 of Algorithm 1.
- 3: Calculate the second PLS component t_2 following the steps:
 - 3.1 Estimate the coefficient a_{2j} of x_j^* in the beta-PLS regression of Y on (T_1, X_j^*) , using the ML method based on (2.7) and (2.18), where $\mu_i = g^{-1}(a_{1j}t_{1i} + a_{2j}x_{ji}^*)$, with $i = 1, \dots, n$, repeating this step for $j = 1, \dots, k$ and so constructing the vector $\underline{a}_2 = (a_{21}, \dots, a_{2k})^\top$.
 - 3.2 Proceed as in steps 3.2 to 3.4 of Algorithm 1.
- 4: State the optimal number $c < k$ of PLS components and calculate the other PLS components t_h , for $h = 3, \dots, c$, following the steps:
 - 4.1 Estimate the coefficient a_{hj} of x_j^* in the beta-PLS regression of Y on $(T_1, \dots, T_{h-1}, X_j^*)$, using the ML method based on (2.7) and (2.18), where $\mu_i = g^{-1}(a_{1j}t_{1i} + \dots + a_{(h-1)j}t_{(h-1)i} + a_{hj}x_{ji}^*)$, with $i = 1, \dots, n$, repeating this step for $j = 1, \dots, c$ and so constructing the vector $\underline{a}_h = (a_{h1}, \dots, a_{hc})^\top$.
 - 4.2 Proceed as in steps 4.2 to 4.4 of Algorithm 1.
- 5: Estimate the loading vector $\underline{q}$ (with no intercept), $\hat{\underline{q}} = (\hat{q}_1, \dots, \hat{q}_c)^\top$ namely, based on (1) the c PLS components $t_1, \dots, t_c$; (2) the beta-PLS model given in (2.18); and (3) the log-likelihood function defined in (2.7).
- 6: Compute the PLS coefficients using latent factors as $\hat{\underline{\beta}}_{\text{PLS}} = \mathbf{W}\hat{\underline{q}}$.
- 7: Predict values for the response Y with the estimated beta PLS model given by $\hat{y}_{\text{pred}} = g^{-1}(\underline{x}_{\text{pred}}^\top \hat{\underline{\beta}})$, where $\underline{x}_{\text{pred}}$ is a vector with the values of the covariates to be used for predicting and $\hat{\underline{\beta}}$ is associated with the PLS coefficients of the model $\hat{y} = g^{-1}(\underline{x}^* \hat{\underline{\beta}}_{\text{PLS}})$ depending on the original variables, after the corresponding algebraic manipulation.

2.4 | Determining the number of PLS components

Akaike (AIC) and Bayesian/Schwarz (BIC) information criteria are based on the entropy concept by means of the log-likelihood function, plus a component penalizing it. AIC and BIC are often used to select statistical models and respectively given, in general, by

$$\text{AIC} = -2\ell(\hat{\underline{\theta}}) + 2k_1, \quad \text{BIC} = -2\ell(\hat{\underline{\theta}}) + k_1 \log(n),$$

where $\ell(\hat{\underline{\theta}})$ is the log-likelihood function of the model with parameter vector $\underline{\theta}$ evaluated at $\underline{\theta} = \hat{\underline{\theta}}$, n is the sample size, and k_1 is the amount of unknown model parameters to be estimated, which on our case is $k_1 = k + 2$. A smaller value of AIC (or BIC) indicates a better model. Li et al¹² mentioned that fixing the number of components in the first local minimum corresponds to the statistical model that best fits the data. The global minimum value should not be selected because the information criteria will always be smaller as the number of components increases, concluding that a better model would have the same number of components and variables; see Osten.³³

To select the number of PLS components, when it is not possible to choose a local minimum value by using some information criterion, the PRESS method is recommended. The PRESS is a cross-validation method frequently used in PLS regression to select the amount of components. The PRESS method assesses the effect of removing a case in different modeling scenarios, that is, doing to vary the number of PLS components. Thus, we proceed until that each of the n cases has been removed, repeating the process until completing the c components. Then, PRESS values should then be compared for the different component numbers to determine the optimum quantity. The PRESS value for the h -th component is computed by

$$\text{PRESS}(h) = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_{i(i)})^2, \quad h = 1, \dots, c, \quad (2.19)$$

where $\hat{Y}_{i(i)}$ is the i -th prediction of Y_i regardless Y_i in the training model, such as a Jackknife estimate. To select the optimal number of PLS components using the PRESS method, a criterion based on the F distribution and proposed by Osten³³ can be used. This is called the Osten F criterion and expressed as

$$F = \frac{(n - (h - 1))(\text{PRESS}(h) - \text{PRESS}(h + 1))}{\text{PRESS}(h + 1)}, \quad (2.20)$$

where $\text{PRESS}(h)$ is defined in (2.19), h is the number of PLS components, and n is the sample size. The statistic given in (2.20) is compared to the $(1 - \alpha) \times 100$ -th percentile of the F distribution denoted by $F_{1-\alpha}(1, n - h + 1)$. Note that the Osten F criterion corresponds to a cutoff point for establishing a benchmark as reference, acting similarly to the Mahalanobis distance, but it is not a hypothesis testing properly. In addition, the Osten F criterion is similar to the Wold R criterion for selecting the optimal number of components in PCA; see Wold.³⁴

3 | DIAGNOSTICS

3.1 | Residual analysis

To assess possible departures from the distributional assumption, as well as detecting atypical observations, several types of residuals have been used in regression models; see Leiva et al³⁵ for different residuals and a study of their adequacy. However, finding residuals whose empirical distribution is close to normality in nonnormal models is not an easy task.

Ferrari and Cribari-Neto²⁶ used a standardized residual for the beta regression model. We consider a type of residual used in generalized additive models for location scale and shape (GAMLSS); see previous studies^{36,37} and a package named `gamlss` of the R software described in Section 4.1. This is the randomized quantile (RQ) residual defined by

$$r_i^{\text{RQ}} = \Phi^{-1} \left(F \left(y_i; g^{-1}(t_i \hat{q}) \hat{\delta}, (1 - g^{-1}(t_i \hat{q})) \hat{\delta} \right) \right), \quad (3.1)$$

where Φ^{-1} is the inverse function of the $N(0, 1)$ cumulative distribution function (CDF) or quantile function, and $F(y; \mu, \delta) = \int_0^y f(u; \mu, \delta) du$ is the Rbeta CDF with PDF $f(y; \mu, \delta)$ being given in (2.1), but based on the reparameterization considered in (2.3). Note that the RQ residual defined in (3.1) must be compared to the normal distribution for evaluating the fitting of the beta PLS model to the data. We propose to use a theoretical probability versus empirical probability (PP) plot to do this evaluation. In addition, observe that the PP plot can be linked to the Kolmogorov-Smirnov (KS) test, by means of which acceptance bands may be constructed inside of this plot. Algorithm 3 summarizes such a construction; see previous studies.³⁸⁻³⁹ Also, in order to confirm the results from the PP plot, we use a theoretical quantile versus empirical quantile (QQ) plot based on the plot of simulated envelopes proposed by Atkinson.⁴⁰

3.2 | Cook distance

The Cook distance is useful to identify atypical cases, measuring the effect on the estimated beta PLS coefficients, $\hat{q}$, namely, by eliminating a case; see Cook.⁴¹ In normal linear regression models, the influence of the case i can be measured by the Cook distance. An equivalent measure of diagnostic to the Cook distance for PLS regression is given by

$$\frac{2}{c} \left(\ell(\hat{\theta}) - \ell(\hat{\theta}_{(i)}) \right), \quad i = 1, \dots, n, \quad (3.2)$$

where c is the optimal number of PLS components, ℓ is the corresponding log-likelihood function, and $\hat{\theta}_{(i)}$ is an estimate of the PLS coefficient vector without considering the case i . According to Williams,⁴² the distance presented in (3.2) can be approximated by means of a type of generalized Cook distance expressed as

$$d_i^{\text{GCook}} \approx \frac{o_i \left(r_i^{\text{RQ}} \right)^2}{c(1 - o_i)}, \quad i = 1, \dots, n, \quad (3.3)$$

where r_i^{RQ} is the RQ residual defined in (3.1) and o_i is the i -th diagonal element of the associated projection matrix. We consider the case i as influential if d_i^{GCook} given in (3.3) is greater than the cutoff point $4/(n - c - 1)$, which is considered as benchmark; for more details about this cutoff point, see Fox.^{43(p.34)}

Algorithm 3 Goodness of fit to any distribution

- 1: Consider data $r_1, \dots, r_n$ and order them as $r_{1:n}, \dots, r_{n:n}$.
 - 2: Estimate the parameter $\underline{\theta}$ of the distribution by $\hat{\underline{\theta}}$ with $r_1, \dots, r_n$ and the ML method.
 - 3: Compute $\hat{v}_{j:n}^* = F(r_{j:n}; \hat{\underline{\theta}})$, for $j = 1, \dots, n$, with F being the corresponding CDF.
 - 4: Calculate $\hat{l}_j = \Phi^{-1}(\hat{v}_{j:n}^*)$.
 - 5: Obtain $\hat{u}_{j:n} = \Phi(\hat{z}_j)$, with $\hat{z}_j = (\hat{l}_j - \bar{l})/s_l$, $s_l = (\sum_{j=1}^n (\hat{l}_j - \bar{l})^2 / (n - 1))^{1/2}$, and $\bar{l} = \sum_{j=1}^n \hat{l}_j / n$.
 - 6: Draw the PP plot with points $w_{j:n} = (j - 0.5)/n$ versus $\hat{u}_{j:n}$, for $j = 1, \dots, n$.
 - 7: Specify a significance level α .
 - 8: Construct acceptance bands according to $(\max\{w - d_{1-\alpha}^{\text{KS}} + 0.5/n, 0\}, \min\{w + d_{1-\alpha}^{\text{KS}} - 0.5/n, 1\})$, where $d_{1-\alpha}^{\text{KS}}$ is the $(1 - \alpha) \times 100$ -th percentile of the distribution (adapted) of the KS distance and w is a continuous version of $w_{j:n}$.
 - 9: Determine the p -value of the KS statistic and reject the null hypothesis of the corresponding distribution for the specified significance level α based on this p -value.
 - 10: Corroborate coherence between steps 8 and 9.
-

3.3 | Mahalanobis distance

The Mahalanobis distance is the length between an observation (case) and the centroid of a multivariate space (global average). This distance is used to identify outliers and given by

$$d_i^{\text{Maha}} = \sqrt{(\underline{X}_i - \bar{\underline{X}})^\top \mathbf{S}_X^{-1} (\underline{X}_i - \bar{\underline{X}})}, \quad i = 1, \dots, n, \quad (3.4)$$

where $\underline{X}_i$ is a random vector, $\bar{\underline{X}}$ is a vector of mean (centroid), and $\mathbf{S}_X$ is the sample variance-covariance matrix of $\mathbf{X}$. Note that it is not possible to use the Mahalanobis distance as a criterion for detecting outliers when there are more variables than cases. Then, its calculation in PLS regression is based on the PLS component matrix, $\mathbf{T}$ namely; see Varmuza and Filzmoser.⁴⁴ Thus, the Mahalanobis distance defined in (3.4) based on c components is expressed as

$$d_i^{\text{Maha}} \approx \sqrt{(\underline{T}_i - \bar{\underline{T}})^\top \mathbf{S}_T^{-1} (\underline{T}_i - \bar{\underline{T}})}, \quad i = 1, \dots, n, \quad (3.5)$$

where $\underline{T}_i$ is a random vector, $\bar{\underline{T}}$ is a vector of mean (centroid), and $\mathbf{S}_T$ is the sample variance-covariance matrix of the PLS components. If d_i^{Maha} is greater than the cutoff point $\sqrt{\chi_{1-\alpha}^2(c)}$, then the case i is atypical; see Varmuza and Filzmoser.⁴⁴ Note that the cutoff point $\chi_{1-\alpha}^2(c)$ is the $(1 - \alpha) \times 100$ -th percentile of the chi-square distribution with c degrees of freedom.

4 | APPLICATION

4.1 | Software and data

R is a noncommercial software for statistical analysis and graphs, with open source code. This software may be secured freely from <http://www.r-project.org>. We implement a computational framework in R to analyze data with the methodology of modeling and diagnostics proposed in the paper. The R codes used in this application are available from the authors upon request. The beta distribution can be symmetric or asymmetric depending on the values of its parameters. We use an adjusted box plot for data following asymmetric distributions, which is implemented in R by the `adjbox` command of the `robustbase` package; see Rousseeuw et al.⁴⁵ The beta PLS regression model is implemented in the `plsRbeta` package and may be performed by a command of the same name; see Bertrand et al.⁴⁶ The `plsRbeta` package is based on the `plsRglm` package, which allows us to estimate the regression parameters and to perform a distributional comparison of the beta PLS model with distributions of the exponential family (for example, the gamma distribution); see Cribari-Neto and Zeileis⁴⁷ and Bertrand et al.⁴⁶ For the diagnostic analysis, the Cook distance is implemented in the `stats` package

through the `cooks.distance` command, which was modified by us for obtaining the generalized Cook distance defined in (3.3). The Mahalanobis distance is implemented in the `chemometrics` package by the `Moutlier` command, which was adapted by us for obtaining the Mahalanobis distance defined in (3.5); see Varmuza and Filzmoser⁴⁴ and Filzmoser and Varmuza.⁴⁸

We have 99 observations of the proportion of kaolinite (Y), which we call “kaolinite data.” These data were collected by an anonymous mining company, for which reflectance was measured through a process with NIR between 350 and 2500 nm. The aim is to predict the proportion of kaolinite based on reflectance measurements corresponding to spectral data (X), which we call “spectral data.”

4.2 | Exploratory data analysis

We perform an exploratory analysis of kaolinite data and their spectra to formulate a suitable model describing them. Figure 1 (left) shows a histogram of the kaolinite proportion. In addition, Figure 1 (right) displays the standard box plot and the mentioned adjusted box plot for asymmetric data. Furthermore, Table 1 presents a descriptive summary, which includes the minimum value $y_{(1)}$, median (MD), mean $\bar{y}$, maximum value $y_{(n)}$, SD, and coefficients of variation (CV), skewness (CS), and kurtosis (CK) for kaolinite data. From Figure 1 and Table 1, note that the data seem to follow a right asymmetric distribution. From the box plots, we observe some outliers. Figure 2 represents 99 spectral data, with each line corresponding to 2151 covariates. Therefore, on the basis of the nature of the phenomenon, amount of covariates, and

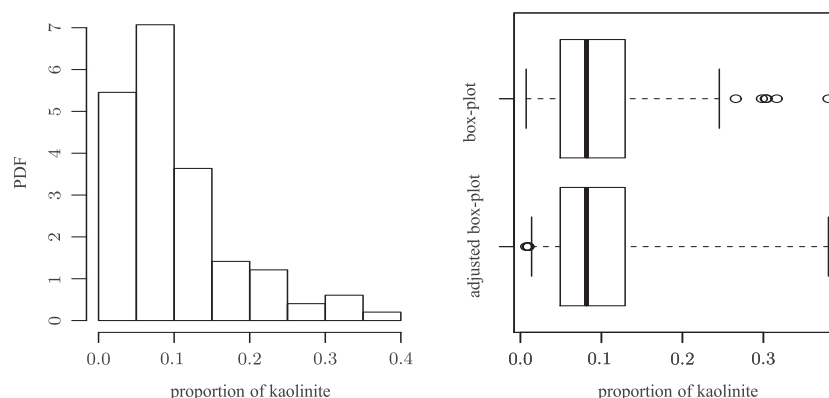


FIGURE 1 Histogram (left) and box plots (right) for kaolinite data. PDF, probability density function

TABLE 1 Descriptive statistics for kaolinite data

n	$y_{(1)}$	MD	$\bar{y}$	$y_{(n)}$	SD	CV	CS	CK
99	0.0071	0.0820	0.1032	0.3803	0.0780	0.7560	1.3382	4.5088

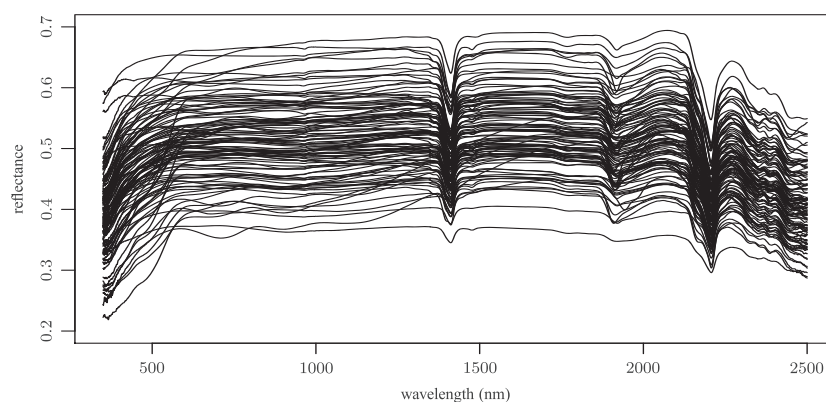


FIGURE 2 Plot of spectra for covariates related to kaolinite data

results from the exploratory data analysis, the beta PLS regression seems to be a suitable model to predict the proportion of kaolinite based on spectral data.

4.3 | Modeling, estimation, and inference

To establish whether the beta PLS model is appropriate to the kaolinite data, we compare it to different models used in the literature. When assuming the normal distribution as part of the GLM, different link functions can be considered. Note that only the case of identity link is equivalent to the standard version of the PLS regression. We compare the following PLS models with the indicated link function:

- Beta PLS model with logit link.
- Beta PLS model with logarithmic link.
- Gamma PLS model with logarithmic link.
- Gamma PLS model with identity link.
- Normal PLS model with identity link.
- Normal PLS model with logarithmic link.

The mentioned link functions are given by

$$g(\mu_i) = \begin{cases} \mu_i, & (\text{identity link}); \\ \log(\mu_i), & (\text{logarithmic link}); \\ \log\left(\frac{\mu_i}{1-\mu_i}\right), & (\text{logit link}). \end{cases}$$

Figure 3 shows AIC (left) and BIC (right) values for different models (from 1 to 30 components) based on the 6 mentioned competing PLS models. From Figure 3, note that the criterion is not conclusive, because no local minimum is detected by the AIC or BIC for the number of components considered. However, observe that the beta PLS model with logit link seems to be the most appropriate one, regardless of the amount of PLS components used for the model. Notice that, for the beta PLS model with logit link, a first local minimum is not detected in Figure 3. AIC and BIC can be used to choose the distribution and link function for modeling the kaolinite proportion, but not for determining the number of PLS components. Then, as mentioned in Section 2.4, we use the PRESS method to obtain the optimal number of components to be used. Selection of the optimal number of PLS components is based on the Osten F criterion defined in (2.20). Figure 4 displays PRESS values for the beta PLS model, where a white circle shows the first value for the mentioned F -statistic, which is greater than the corresponding cutoff point. This suggests a beta PLS model with $c = 15$ components.

On the basis of our exploratory data analysis, we assume that the response $Y_i \sim \text{Rbeta}(\mu_i, \delta)$ for kaolinite data, whose systematic component of the corresponding regression model for the mean, by using the logit link function, is expressed as

$$E[Y_i] = \mu_i = g^{-1}(\eta_i) = g^{-1}(\mathbf{x}_i^T \underline{\beta}), \quad i = 1, \dots, 99, \quad (4.1)$$

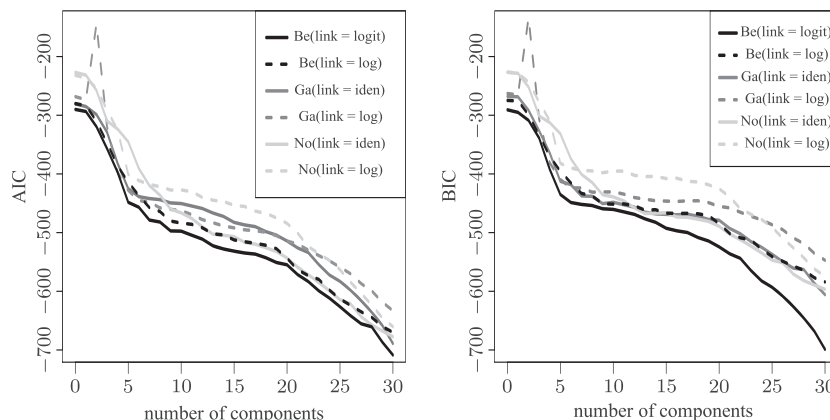


FIGURE 3 Number of components versus Akaike information criterion (AIC) (left) and Bayesian/Schwarz information criterion (BIC) (right) values for the indicated partial least squares model and link function with kaolinite data

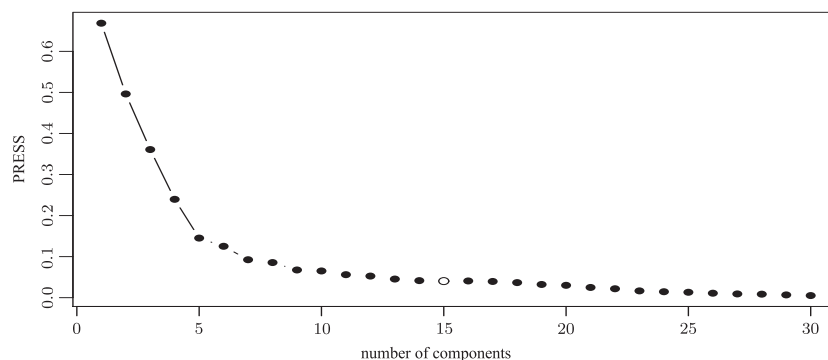


FIGURE 4 Number of components versus prediction error sum of squared (PRESS) values for the beta partial least squares model with kaolinite data

TABLE 2 Relative change (RC) of the indicated estimated parameter after dropping the indicated case(s) for kaolinite data

θ	None		{#33}				{#78}				{#33, #78}			
	$\hat{\theta}$	<i>p</i> -value	$\hat{\theta}$	<i>p</i> -value	$RC_{\hat{\theta}_{(0)}}$	$RC_{SE(\hat{\theta}_{(0)})}$	$\hat{\theta}$	<i>p</i> -value	$RC_{\hat{\theta}_{(0)}}$	$RC_{SE(\hat{\theta}_{(0)})}$	$\hat{\theta}$	<i>p</i> -value	$RC_{\hat{\theta}_{(0)}}$	$RC_{SE(\hat{\theta}_{(0)})}$
q_1	0.007	<.01	0.007	<.01	1.140	0.758	0.008	<.01	8.458	4.987	0.008	<.01	8.705	3.220
q_2	0.047	<.01	0.047	<.01	1.601	0.647	0.050	<.01	7.154	4.518	0.051	<.01	8.951	4.710
q_3	0.175	<.01	0.185	<.01	5.594	6.146	0.209	<.01	19.297	3.408	0.226	<.01	28.938	12.680
q_4	0.185	<.01	0.181	<.01	2.203	0.335	0.156	<.01	15.347	15.684	0.153	<.01	17.290	15.331
q_5	0.202	<.01	0.206	<.01	2.184	2.518	0.162	<.01	19.949	0.773	0.160	<.01	20.646	0.076
q_6	0.197	<.01	0.204	<.01	3.404	0.725	0.136	<.01	31.271	13.353	0.143	<.01	27.732	13.148
q_7	0.248	<.01	0.246	<.01	0.720	0.803	0.161	.001	35.199	11.321	0.164	.001	33.680	10.572
q_8	0.397	<.01	0.417	<.01	5.077	3.930	0.487	<.01	22.627	22.406	0.568	<.01	42.836	25.178
q_9	0.154	.015	0.208	.004	35.376	13.625	0.122	.043	20.911	0.420	0.125	.040	19.004	3.835
q_{10}	0.226	.015	0.227	.014	0.409	0.528	0.373	.003	65.215	36.829	0.412	.001	82.419	33.981
q_{11}	0.767	<.01	0.672	<.01	12.312	14.573	0.373	.018	51.331	28.499	0.548	<.01	28.513	37.979
q_{12}	0.759	<.01	0.734	<.01	3.272	16.075	0.310	.023	59.123	36.450	0.282	.013	62.894	47.522
q_{13}	0.743	.001	0.980	<.01	31.916	18.893	0.450	.005	39.442	28.441	0.526	.015	29.182	2.674
q_{14}	0.909	.003	0.706	.013	22.363	7.178	0.895	.002	1.530	7.923	0.906	.005	0.372	4.731
q_{15}	1.073	.002	0.804	.007	25.133	14.049	1.283	<.01	19.490	12.821	0.964	<.01	10.187	21.486
δ	227.590	<.01	288.564	<.01	0.421	0.941	272.297	<.01	20.199	19.644	270.033	<.01	18.649	19.821

where $x_i = (1, x_{i1}, \dots, x_{i2151})^T$ are the observations of 2151 covariates (plus a “one” in the first position related to the constant term of regression), $\beta = (\beta_0, \beta_1, \dots, \beta_{2151})^T$ and $g(\mu) = \log(\mu/(1 - \mu))$. The model defined in (4.1) can be written as a beta PLS model by

$$\begin{aligned} E[Y_i] &= g^{-1}(t_i^T \underline{q}) = g^{-1}(q_1 t_{i1} + \dots + q_{15} t_{i15}) \\ &= g^{-1}(\underline{x}_i^T \underline{\beta}) = g^{-1}(\beta_0 + \beta_1 x_{i1} + \dots + \beta_{2151} x_{i2151}), \quad i = 1, \dots, 99, \end{aligned} \quad (4.2)$$

where $t_{i1}, \dots, t_{i15}$ are obtained from the PLS components computed with Algorithm 2. The ML estimates of the PLS coefficients given in (4.2), and *p*-values of the corresponding *t*-tests, are presented in Table 2. Note that all coefficients are significant at 5%. From the estimated coefficients provided in Table 2 for the model defined in (4.2), we can obtain the ML estimates $\hat{\underline{\beta}}$. Because of restrictions of space, we show graphically the values of the 2152 coefficients (including β_0) associated with $\hat{\underline{\beta}}$ in Figure 5.

4.4 | Diagnostics and model checking

Model assumptions given in (4.1) and (4.2) are verified by using the RQ residual for kaolinite data. Figure 6 (left) shows this residual, from which no unusual features are detected. Figure 6 (right) displays a graphical plot of observed values versus predicted values of the kaolinite proportion, which shows a good agreement between both values. Figure 7 (left)

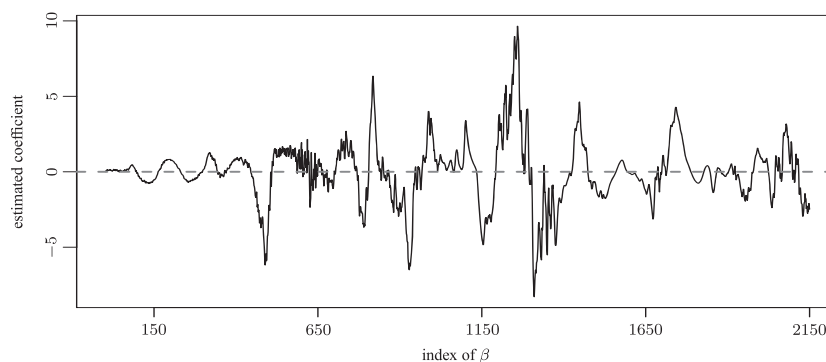


FIGURE 5 Index plot of the maximum likelihood estimates of partial least squares β coefficients for kaolinite data

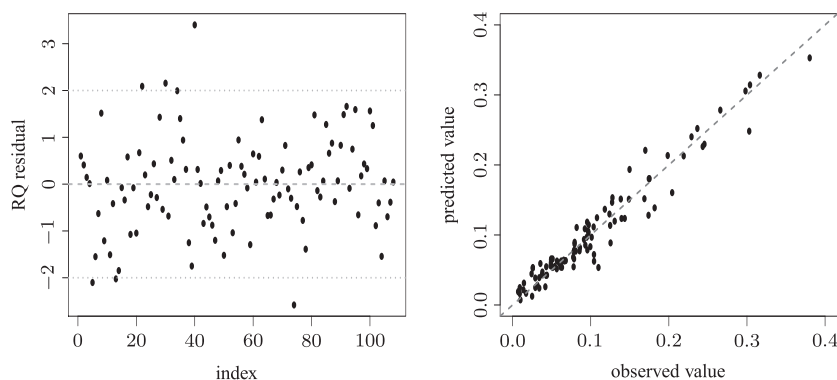


FIGURE 6 Index plot of the randomized quantile (RQ) residual (left) and predicted versus observed values (right) for kaolinite data

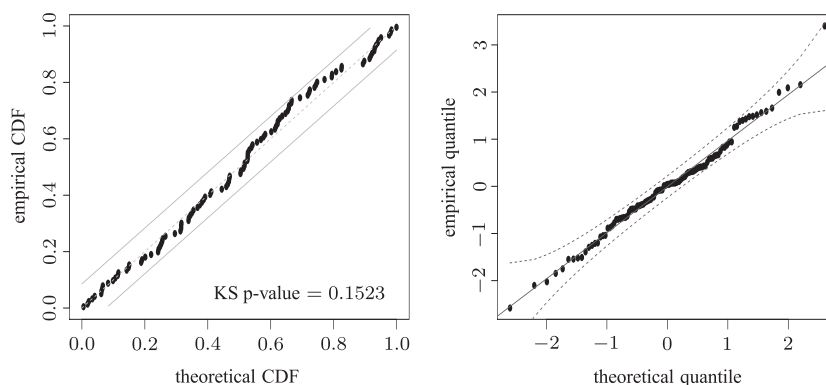


FIGURE 7 PP plot with 95% acceptance bands (left) and QQ plot with its simulated envelope (right) for RQ residuals using kaolinite data. CDF, cumulative distribution function

sketches a PP plot with 95% acceptance bands to verify the distributional assumption of the model given in (4.1) and (4.2). Figure 7 (right) presents a QQ plot with simulated envelope to evaluate the normality fit to the RQ residuals. Note that KS p -value = 0.1523, which supports to a 5% of significance the normality assumption of the RQ residuals obtained from the beta PLS regression. This figure does not show unusual features, so that the assumption of a response R_{β} distributed seems to be suitable.

Diagnostics based on the generalized Cook and Mahalanobis distances for the beta PLS model given in (4.1) and (4.2) are presented in Figure 8. The index plot of the generalized Cook distance displayed in Figure 8 (left) detects the cases #20, #33, #61, and #78 as atypical and could be influential. Figure 8 (right) provides the index plot of the Mahalanobis distance, from which the cases #33, #78, and #92 are detected as atypical and then potentially influential. Note that if a spectrum has a different trend to another, then different minerals could be present in the rock sample. Figures 9 and

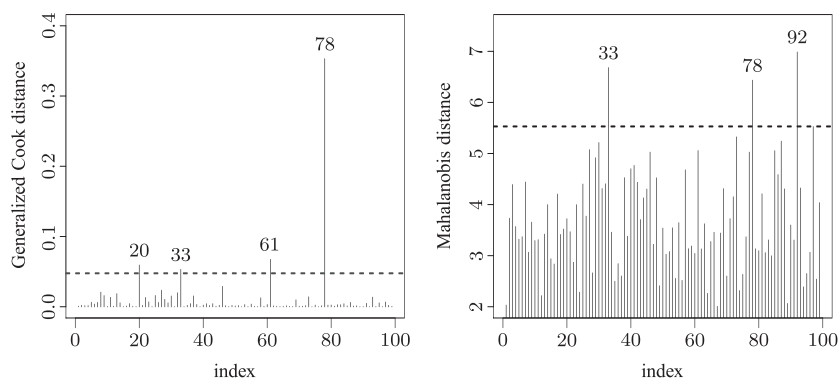


FIGURE 8 Index plots of generalized Cook (left) and Mahalanobis (right) distances for kaolinite data

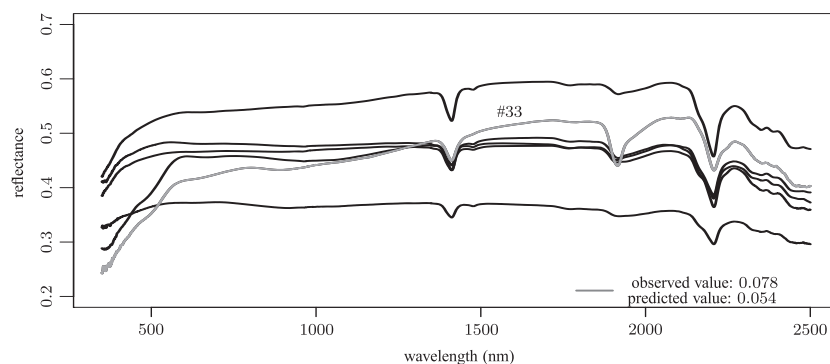


FIGURE 9 Plot of the case #33 (in gray) compared to other spectra with similar kaolinite proportion

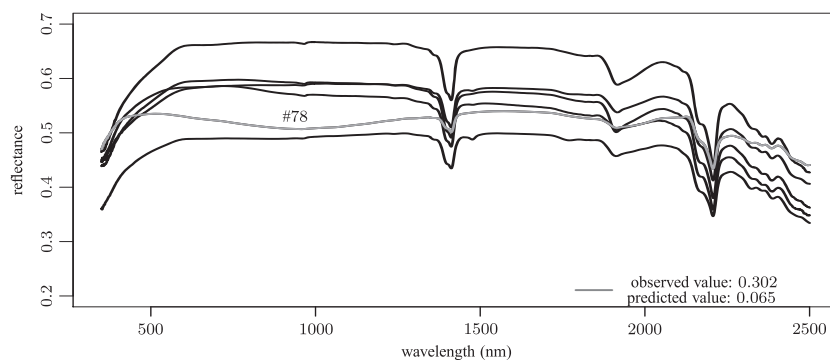


FIGURE 10 Plot of the case #78 (in gray) compared to other spectra with similar kaolinite proportion

10 show the cases (spectra) #33 and #78, respectively, compared to those spectra with a similar proportion of kaolinite. Observe that both cases #33 and #78 display a different trend from the other cases.

We now investigate the impact on the model inference when the detected cases, in both generalized Cook and Mahalanobis distances, are removed. Then, we reestimate the model parameters after removing the sets of cases $\{ \#33 \}$, $\{ \#78 \}$, and $\{ \#33, \#78 \}$. Table 2 provides the relative changes (RCs) in the parameter estimates and in their corresponding estimated SEs, by using the kaolinite data. These RCs are calculated as

$$RC_{\hat{\theta}_{j(i)}} = \left| \frac{\hat{\theta}_j - \hat{\theta}_{j(i)}}{\hat{\theta}_j} \right| \times 100\%, \quad RC_{SE[\hat{\theta}_j]_{(i)}} = \left| \frac{\widehat{SE}[\hat{\theta}_j] - \widehat{SE}[\hat{\theta}_j]_{(i)}}{\widehat{SE}[\hat{\theta}_j]} \right| \times 100\%,$$

where $\hat{\theta}_{j(i)}$ and $\widehat{SE}[\hat{\theta}_j]_{(i)}$ denote the ML estimates of θ_j and of the SE for the associated estimator, respectively, obtained after removing the case i , for $i = 1, \dots, 99$ and $j = 1, \dots, 16$ (15 PLS components plus the shape parameter, δ). From Table 2, note that the cases $\{ \#33 \}$ and $\{ \#33 \text{ and } \#78 \}$ (together) change more than 50% for some RCs in q_{10} and q_{12} . In

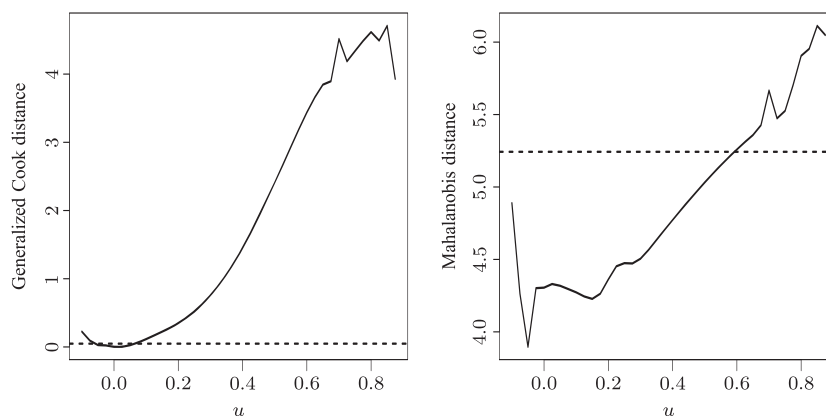


FIGURE 11 Influence diagnostics for the case #88 (perturbed) with kaolinite data

addition, all coefficients are significant to 5% and no inferential changes are detected, with the SE being obtained from expression (2.10), evaluating θ at $\hat{\theta}$. The results presented in Table 2 provide evidence that the diagnostic methods derived in this research identify potentially influential points, but these do not affect the inference of the model. In summary, the diagnostic analysis based on the generalized Cook and Mahalanobis distances, and in the RQ residual, confirm that the beta PLS regression model presented in (4.1) and (4.2) is not sensitive to the detected atypical cases and quite suitable for modeling the kaolinite data.

4.5 | Simulation study

To show the capacity for detecting atypical cases of the diagnostic methods proposed in this paper, we perform a simulation study based on kaolinite data. We perturb a case for kaolinite data corresponding to the value closest to the average of the proportion of kaolinite. The case #88 is considered, whose value is $y_{88} = 0.1037242$. This value is perturbed as follows: $y_{88}^* = y_{88} + u$, with u varying between -0.1 and 0.875 by 0.025 . Our simulation study consists of estimating the beta PLS model parameters using kaolinite data with the original vector $\underline{Y}$ being modified in its case #88 by y_{88}^* . For each value of u , we obtain the generalized Cook and Mahalanobis distances, d_{88}^{GCook} and d_{88}^{Maha} , obtained from expressions (3.3) and (3.5), respectively. Figure 11 shows the values of the generalized Cook (left) and Mahalanobis (right) distances for the perturbed case y_{88}^* in each u . Note that, as the perturbed value moves to its true value, the diagnostic measures tend to increasing until it exceeds the cutoff of the corresponding distance, detecting it as atypical. Furthermore, note that the generalized Cook distance is more sensitive to detect true atypical values than the Mahalanobis distance for kaolinite data. Therefore, this simulation study demonstrates that the generalized Cook and Mahalanobis distances work suitably for identifying atypical cases in the beta PLS regression model.

5 | CONCLUSIONS

The beta PLS regression model has proven to be quite appropriate for data modeling whose response corresponds to a proportion. This is coherent because the support of the response variable is in the range zero to one. Then, using distributions with support in the real line or the positive real line for proportions is not suitable. Thus, considering distributions different to the beta distribution to model proportions could lead to predictions outside the natural range of a proportion, that is, negative predictions or greater than one.

We have considered RQ residuals for the beta PLS regression model, which are commonly used in GAMLSS. Furthermore, we have considered the generalized Cook and Mahalanobis distances to detect influential cases and outliers, respectively, as global diagnostic methods. A simulation study has been conducted to assess the performance of these tools considered. The simulation study has shown their good performance, with the generalized Cook distance being more sensitive to detect true atypical values than the Mahalanobis distance with kaolinite data.

We have modeled data from the real-work related to the proportion of kaolinite, which are frequently analyzed in mining. The beta PLS regression model has been compared to other competitive PLS models, showing a better performance and closer to reality for this kind of data. The results obtained for the beta PLS regression model and its diagnostics may be of interest for the mining industry.

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